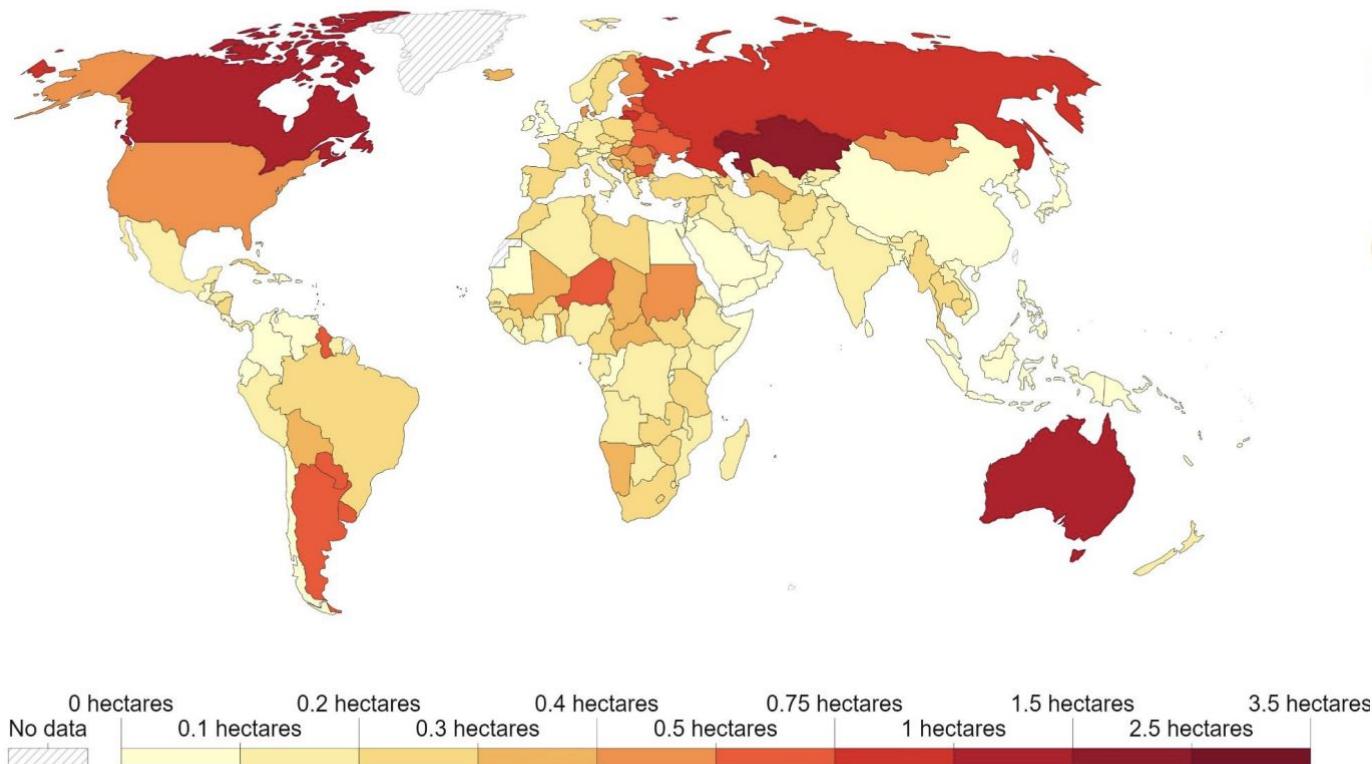


2.1 Population Distribution

Arable land use per person, 2020

Arable land is defined by the FAO as land under temporary crops, temporary meadows for mowing or for pasture, land under market or kitchen gardens, and land temporarily fallow. It is measured in hectares per person.

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- Identify the scale of analysis of the map.
- Identify the type of density shown in the map.
- Explain how the map reveals information about the pressure the population exerts on the land.



2.4.1 Population Dynamics

Objective and Essential Learning

2.4 Explain factors that account for contemporary and historical trends in population growth & decline.

- Demographic factors that determine a population's growth and decline are fertility, mortality and migration.
- Geographers use the rate of natural increase and population doubling time to explain population growth and decline.



What factors determine a population's growth & decline?

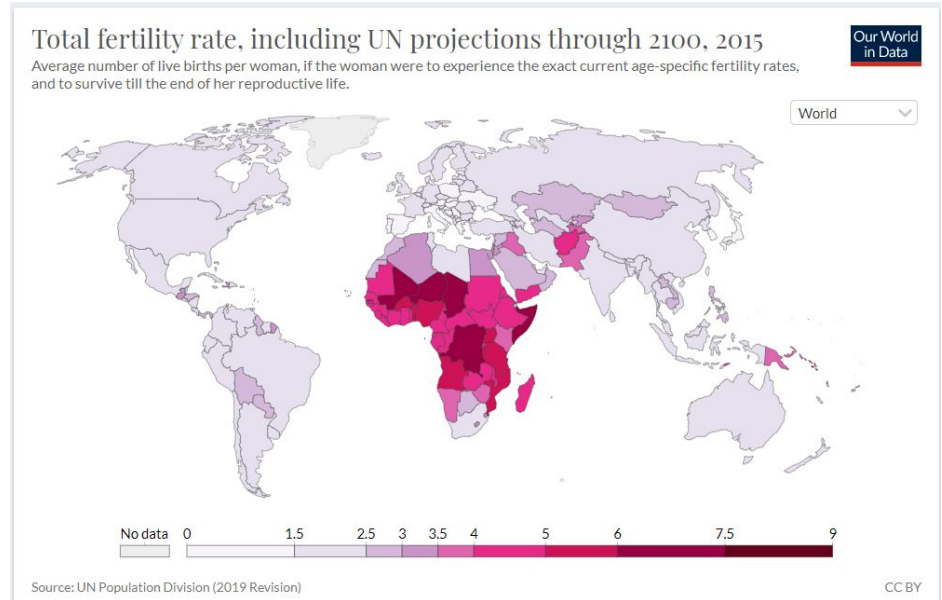
1. Fertility

- a. Crude Birth Rate (CBR)
- b. Total Fertility Rate (TFR)

2. Mortality

- a. Life Expectancy
- b. Crude Death Rate (CDR)
- c. Infant Mortality Rate (IMR)

3. Migration



Fertility

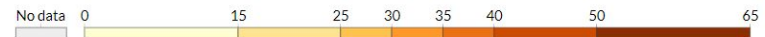
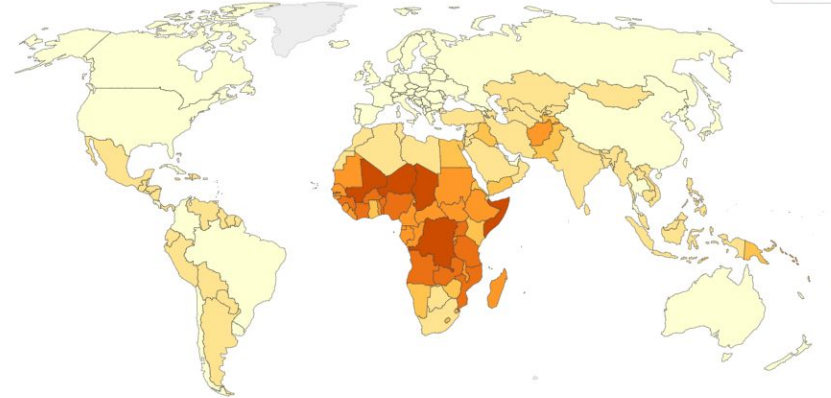
- a. Crude Birth Rate (CBR)
 - i. The number of live births occurring in one year per 1,000 people.
- b. Total Fertility Rate (TFR)
 - i. Average number of children who would be born per woman during her childbearing years.
 - ii. Globally TFR has been significantly decreasing over time.

Birth rate, 2020

The number of live births occurring during the year, per 1,000 people.

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World



Source: United Nations – Population Division (2019 Revision)

CC BY

Mortality

a. Life Expectancy

- i. The number of years the average person will live.
- ii. This has been steady increasing over time, which has contributed to population growth.

b. Crude Death Rate (CDR)

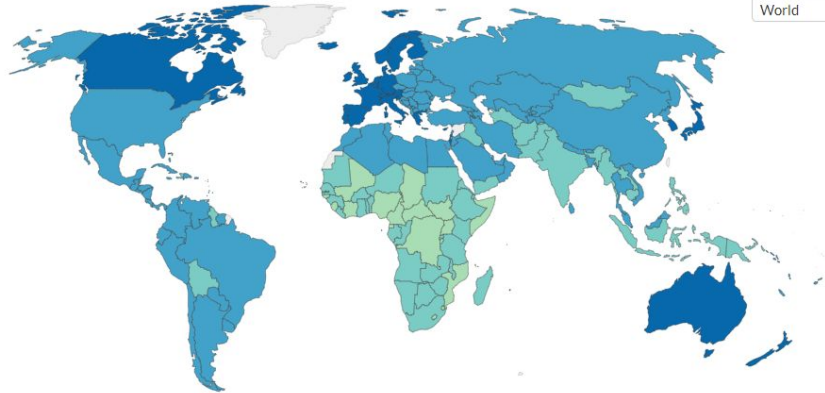
- i. The number of deaths occurring in one year per 1,000 people.

Life expectancy, 2016

The average number of years a newborn would live if age-specific mortality rates in the current year were to stay the same throughout its life.

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World



Source: UN Population Division; World Bank

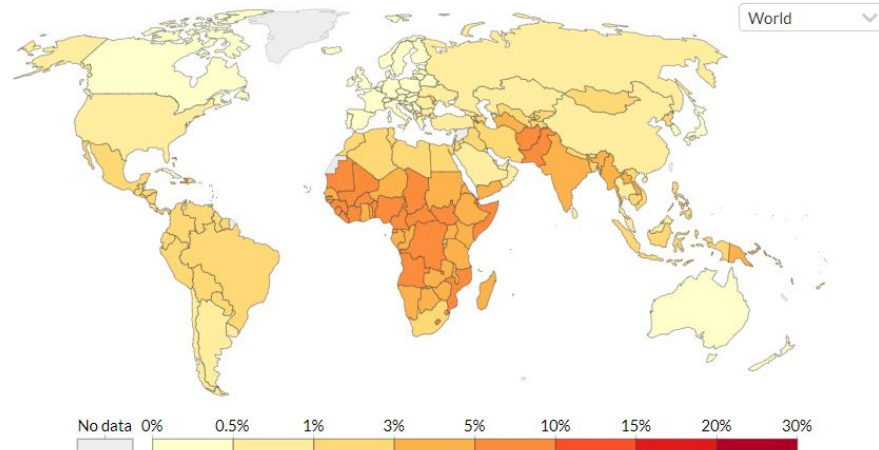
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Mortality

- c. Infant Mortality Rate (IMR)
- i. The number of children who die before one year of age.
 - ii. Typically when IMR goes down, life expectancy goes up which leads to higher population growth.

Infant mortality rate, 2017

The share of newborns who die before reaching one year of age.



Source: UN Interagency Group for Child Mortality Estimation (IGME)

OurWorldInData.org/child-mortality/ • CC BY

How do geographers calculate population growth & decline?

- a. Natural Increase Rate (NIR) or Rate of Natural Increase (RNI)
 - i. Does not account for migration.
 - ii. Can be a negative number, indicating population decrease.
 - iii. Usually expressed in a percentage

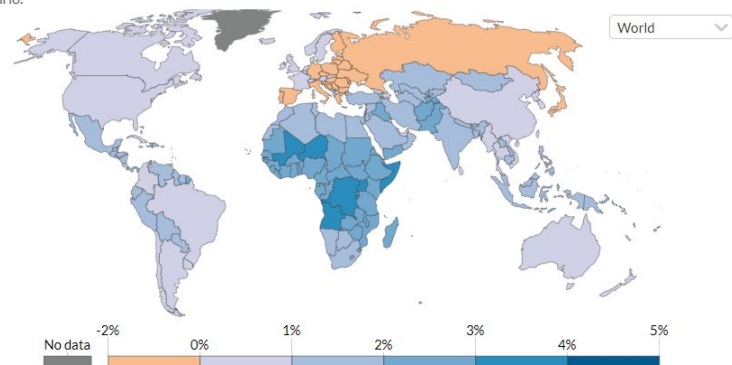
CBR - CDR

10

Natural population growth, 2020

Natural population growth is the population increase determined by births and deaths. Migration flows are not taken into account. This is shown from 1950, with UN projections to 2099 based on its median scenario.

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Source: United Nations - Population Division (2019 Revision)

OurWorldInData.org/world-population-growth/ • CC BY

How do geographers calculate population growth & decline?

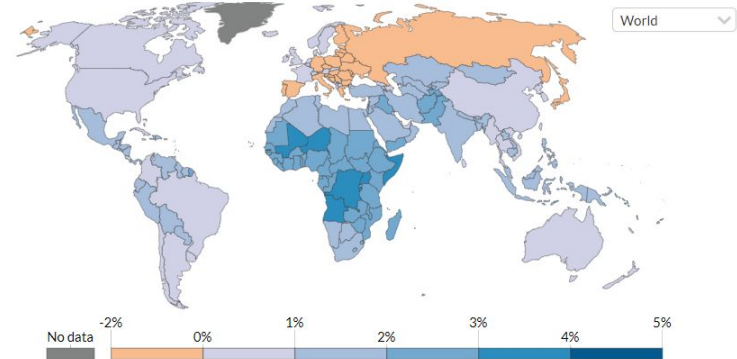
- a. Natural Increase Rate (NIR) or Rate of Natural Increase (RNI)
 - i. NIR - United States
 - ii. CBR - 12.4
 - iii. CDR - 8.3

$$\frac{12.4 - 8.3}{10} = 0.41\%$$

Natural population growth, 2020

Natural population growth is the population increase determined by births and deaths. Migration flows are not taken into account. This is shown from 1950, with UN projections to 2099 based on its median scenario.

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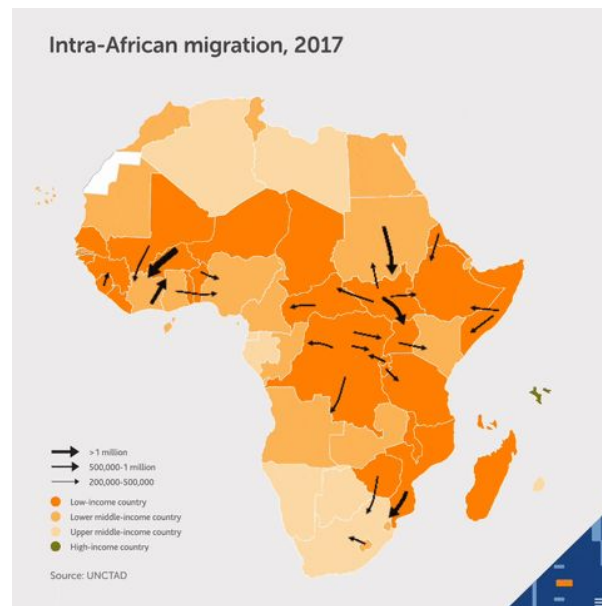
Source: United Nations - Population Division (2019 Revision)

OurWorldInData.org/world-population-growth/ • CC BY

How do geographers calculate population growth & decline?

b. Population Growth Rate

- i. More accurate measure because it accounts for migration.
- ii. Immigrants - people who moved into a country
- iii. Emigrants - people that move out of the country.
- iv. United States - 0.72%



$$\text{Births} - \text{Deaths} + \text{Immigrants} - \text{Emigrants}$$

How do geographers calculate population growth & decline?

- c. Doubling Time: The number of years in which a population will double assuming the growth rate remains stable.

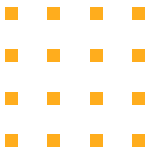
$$\frac{70}{\text{NIR}} \quad \bigg| \quad \frac{70}{0.41\%} = 175 \text{ years}$$

Objective and Essential Learning

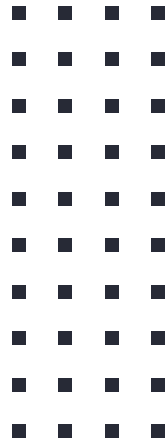
2.4 Explain factors that account for contemporary and historical trends in population growth & decline.

- Demographic factors that determine a population's growth and decline are fertility, mortality and migration.
- Geographers use the rate of natural increase and population doubling time to explain population growth and decline.





2.4.2 Population Dynamics



Objective and Essential Learning

2.4 Explain factors that account for contemporary and historical trends in population growth & decline.

- Social, cultural, political, and economic factors influence fertility, mortality and migration rates.

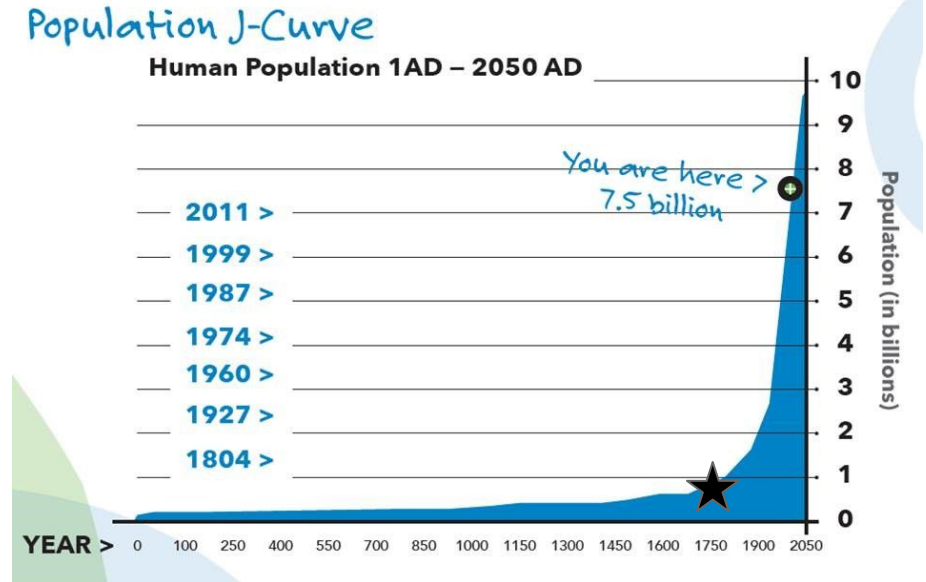


2.8 Explain how the changing role of females has demographic consequences in different parts of the world.

- Changing social values and access to education, employment, health care, and contraception have reduced fertility rates in most parts of the world.
- Changing social, economic, and political roles for females have influenced patterns of fertility, mortality....

Historical Trends in Population Growth & Decline

- Throughout most of history, population growth was very low.
- Starting around 1750
 - Agricultural Revolution
 - Industrial Revolution
 - Rapid urbanization
 - Advances in healthcare & sanitation
- First 1 billion in 1804
 - Since then, doubling time has decreased as population growth increased rapidly



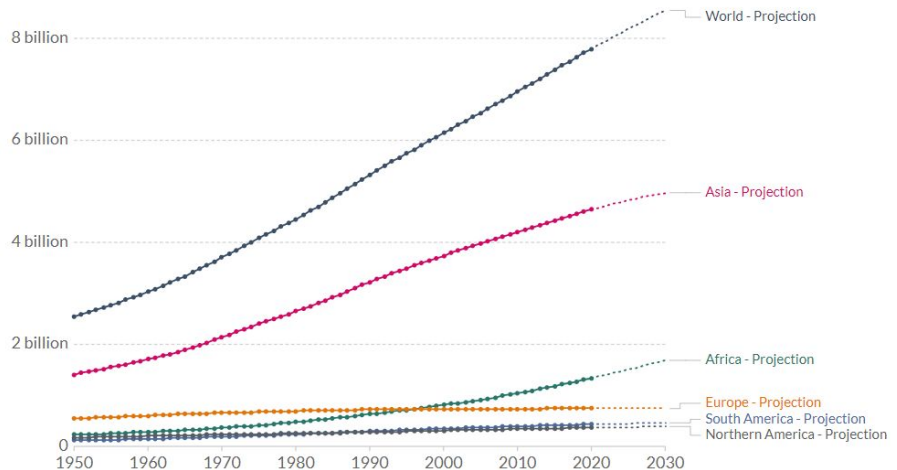
Source: <https://www.populationconnection.org/talking-about-population-growth/>

Contemporary Trends in Population Growth & Decline

- Today, most population growth is happening in LDCs - specifically in Asia and Africa.
 - NIR in Niger = 3.8%
 - NIR in Spain = -0.11%

Population of all world regions, including the UN projection until 2100

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Source: United Nations - Population Division (2019 Revision)

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What causes population to grow & decline?



ECONOMIC

Population Growth - High TFR, CBR or life expectancy // Low CDR & IMR

- Agricultural societies
 - Need children to labor on the farm
- Advancements in food production & nutrition
 - Mechanization of agriculture
 - Farming efficiency
- Advancements in sanitation
 - Less water contamination & disease
 - Water treatment plants
 - Garbage pickup
- Economic prosperity
- Access to healthcare
 - Core countries provide better healthcare services
 - Prevention & cure of disease.
 - Women have access to pre & post-natal care

What causes population to grow & decline?



ECONOMIC

Population Decline - High CDR & IMR // Low TFR, CBR, or life expectancy

- Industrial & post-industrial society
 - Women more involved in the workforce
 - Urbanization -> small living quarters
- Economic hardship
 - Children are expensive, provision of food & resources
- Access to healthcare
 - Women have access to contraceptives

What causes population to grow & decline?



SOCIAL

Population Growth - High TFR, CBR or life expectancy // Low CDR & IMR

- Religion
 - Traditional cultures encourage big families
 - Restriction of contraceptive use and abortion.
 - Earlier marriages = more children

Population Decline - High CDR & IMR // Low TFR, CBR, or life expectancy

- Role of Women - Education & Workforce
 - Cultural expectations have changed in many MDCs -> more women in the workforce/education -> postpone having children -> reduction in child bearing years -> less children.
- Healthcare
 - How to care for babies -> lower IMR, CDR, higher life expectancy
 - Access & understanding of contraception -> lower TFR

What causes population to grow & decline?



POLITICAL

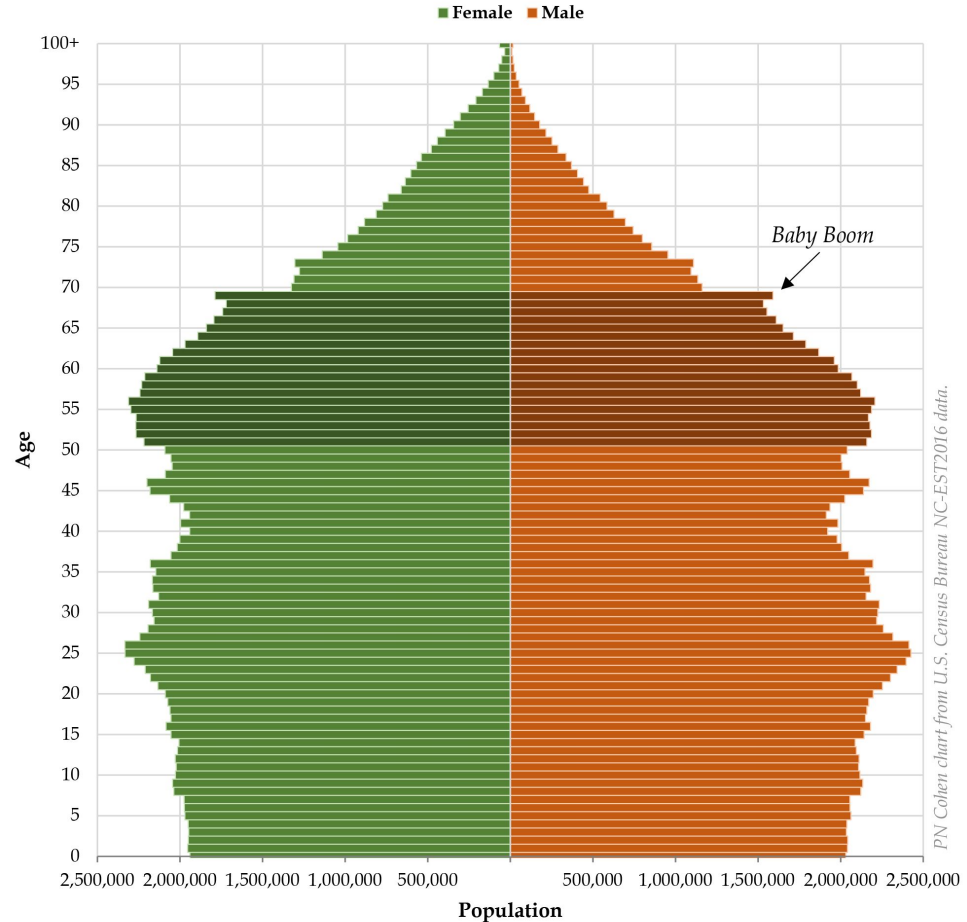
Population Growth - High TFR, CBR or life expectancy // Low CDR & IMR

- Pro-Natalist Population Policies
 - Government encourages families through propaganda and incentives to have children.
- Shift from war to peace -> Baby Boom

Population Decline - High CDR & IMR // Low TFR, CBR, or life expectancy

- Anti-Natalist Population Policies
 - Government discourages families through propaganda, disincentives, and policies.
- War
 - Higher mortality rate,
 - increase in migration
 - shortages of supplies

U.S. Population Pyramid: July 1, 2016



United States "Baby Boom" after World War II (1946-1965)

- End of WWII
- Economic Prosperity
- Cultural Expectations of Women

What causes population to grow & decline?



NATURAL

Population Growth - High TFR, CBR or life expectancy // Low CDR & IMR

Population Decline - High CDR & IMR // Low TFR, CBR, or life expectancy

- Natural disasters (earthquakes, flooding, tsunamis, hurricanes)
- Famine & Drought
- Spread of disease
 - 60% of Europe died in the 1300s due to Bubonic Plague

Objective and Essential Learning

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- Geographers use the rate of natural increase and population doubling time to explain population growth and decline.
- Social, cultural, political, and economic factors influence fertility, mortality and migration rates.

